Replication of "'In- Group Love' and 'Out- Group Hate' as Motives for Individual Participation in Intergroup Conflict: A New Game Paradigm" by N Halevy, G Bornstein, L Sagiv (2008, *Psychological Science*)

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In their original paper, "'Ingroup Love' and 'Outgroup Hate' as Motives for Individual Participation in Intergroup Conflict: A New Game Paradigm," Halevy, Bornstein and Sagiv (2008) discuss why participants follow certain behaviour patterns that negatively impact on themselves during an intergroup conflict. We chose to replicate this particular study because it is close to the first author's research interests in social psychological applications of economic game paradigms (e.g. Thomae, Zeitlyn, Griffiths & Van Vugt, 2013) but also because it fits well into both authors' broader area of expertise in group processes and intergroup behaviour.

Halevy et al. implemented both the intergroup prisoner's dilemma (IPD) game and the intergroup prisoner's dilemma-maximizing difference (IPD-MD) game to distinguish whether participants are seeking to harm the outgroup or help the ingroup in engaging in such "self-sacrificial behaviour." Harm during intergroup conflict in this study is operationalised as allocating tokens to a pool which will not only financially benefit one's own group (ingroup) but simultaneously financially penalise the other experimental group (outgroup).

The IPD game asks participants to either help their ingroup while simultaneously harming the outgroup or to take no action at all for personal gain. Participants in this game can contribute experimental tokens to Pool B, which adds £0.60 per token to the ingroup pool (£0.20 for each player), while simultaneously reducing the outgroup pool by £0.60 (£0.20 from each player) per token. The IPD-MD game, on the other hand, allows participants to make choices that either help their ingroup without harming the outgroup, help their ingroup while simultaneously harming the outgroup, or to take no action at all for personal gain. In the IPD-MD game, participants can either allocate tokens to Pool B, with the same consequences as described above, or can alternatively allocate tokens to Pool A, which adds £0.60 per token to the ingroup pool (£0.20 for each player), without reducing the payoff for the outgroup. Both conditions also allow for 'selfish' behaviour, such that participants can keep any number of tokens, which in this case will neither contribute to the ingroup pool nor deduct from the outgroup pool but will pay £0.40 per token for the individual player.

In this replication of Halevy et al. (2008), we sought to exactly reproduce the methodology used in the original paper. The authors tested whether the IPD-MD game would reveal a specific, unique motivation for behaviour compared to the IPD game. Additionally, they tested whether intragroup discussion would affect the significance of the motives. We focus on their finding that

"the contribution rate to pool B [between group] in the IPD-MD game was significantly lower than the contribution rate to pool B (the only pool) in the IPD game, $F(1, 76) = 109.74$, $p = .000$." (Halevy et al., p. 408)

Method

Power Analysis

The replication team reviewed the article and selected the following effect as the key finding for replication:

> "The most important finding concerns how contributions were divided between pools W [within-group] and B [between-group] in the IPD-MD game. As Figure 1 shows, contributions were made almost exclusively to the cooperative, within group pool. In the no-communication condition, participants contributed an average of 47% of their endowment to pool W and less than 6% to pool B.3 Following within group communication, contributions to pool W increased to 68%, $t(29) = -2.756$, $p = .010$, whereas contributions to pool B remained low, at 4%." (Halevy et al., p. 408)

However, during initial review of the replication study, the original first author suggested that the effect reported in footnote 3 was the key result from that study:

> "The contribution rate to pool B in the IPD-MD game was significantly lower than the contribution rate to pool B (the only pool) in the IPD game, $F(1, 76) = 109.74$, $p = .000$." (Halevy et al., p. 408)

The original study author provided the following rationale:

> "The reason that the high contribution rates to the within group pool in the IPD-MD game are identified as 'the most important finding' is because they reduce intergroup conflict. A direct measure of the degree to which intergroup conflict was reduced appears in footnote 3 (i.e., the significantly lower contribution rates to the harmful between-group pool in the IPD-MD game as compared with the IPD game)." (Correspondence with Dr. Halevy)

Thus, at the author's suggestion, we selected the latter effect as the primary test, and the effect identified by the replication team was identified as a secondary test.

To calculate the sample size necessary for an adequately powered replication, an effect size of partial $\eta^2 = .59$ was determined using Daniel Lakens's effect size calculation table. This was then entered into G*Power, with the test family "F tests" and the statistical test "ANOVA: fixed effects, main effects and interactions" selected. The numerator df was entered as 1 and the number of groups as 2; the effect of interest was the main effect of game type on B pool

contributions. This produced a result of 15 groups being necessary for a replication with 99% power, or 45 participants.

Planned Sample

Despite the power analysis yielding 15 groups as sufficient for 99% power, the present authors planned to collect 40 groups of participants. Groups were three people each for a total of 120 participants.

This sample size was chosen because the suggested 15 was significantly lower than the original paper's 80 groups. Increasing the sample size to 40 groups would lead to more stable means and variances while still being conscious of the limitations of the small participant pool at University of Winchester.

Prior to data collection, it was decided that should 40 groups be collected with more participants remaining in the participant pool and funds still allowing for continued collection, more groups would be included until resources ran out.

Each group of three would be made up of participants of the same sex (ie. 3 females or 3 males).

Participants were recruited through a university participant pool, which also advertised remuneration. All participants were paid, depending on the outcome of the game, a sum between £2 and £14.

Materials

Below are rules for the IPD and IPD-MD games. Participant instructions can be found in the Files section of the Materials component on the OSF. The use of materials below was followed precisely:

"The game is played by two groups, with 3 members in each group. Each player receives an endowment of 10 tokens and can contribute any number of these tokens to the group's pool. For each token contributed by a member of the ingroup, each of its members, including the contributor, gains 1 money unit (MU) and each member of the outgroup loses 1 MU. For each token kept, the player is paid 2 MU. Because the individual's return from contributing a token is 1 MU, but his or her cost is 2 MU, the dominant individual strategy —the strategy that yields the highest personal payoffs regardless of what all the other ingroup and outgroup members do—is to contribute nothing (i.e., defect). However, because contributing a token generates a total of 3 MU for the group while costing only 2 MU, the dominant group strategy—the strategy that yields the highest payoffs for each group regardless of what the other group does—is for all group members to contribute all their tokens."

“To disentangle any motivational ambiguities, we introduce a new paradigm, called the intergroup prisoner’s dilemma–maximizing difference (IPD-MD) game. Like the IPD game, the IPD-MD game involves a competition between two 3-member groups. Each group member receives an endowment of 10 tokens, each worth 2 MU, and can decide how many of these tokens to contribute. Unlike in the IPD game, however, contributions can be made to two different pools. Contributing a token to the within group pool (pool W) increases the payoff for each ingroup member, including the contributor, by 1 MU, without affecting the outgroup. Contributing a token to the between-group pool (pool B) increases the payoff for each ingroup member, including the contributor, by 1 MU, and at the same time decreases the payoff for each outgroup member by 1 MU. The choice between pools W and B is what reveals the specific motivation underlying each individual’s behaviour.” Figure 1 depicts a visual display of the 2 x 2 experimental design.

		Condition: Communication	
		No communication	Communication
Condition: Type of Game	IPD	**Condition 1:** *IPD* *No communication*	**Condition 2:** *IPD* *Within group communication*
	IPD-MD	**Condition 3:** *IPD-MD* *No communication*	**Condition 4:** *IPD-MD* *Within group communication*

Figure 1: Experimental design (2 (Type of Game) by 2 (Communication).

Procedure

The procedure below was followed precisely.

“Participants arrived at the laboratory in cohorts of 6 and were randomly assigned to one of the four conditions (IPD or IPD-MD with or without intragroup communication). The participants in each cohort were randomly divided into two 3-person groups.”

The 6 participants were all of the same sex; there were either two male groups or two female groups.

Each group was escorted into separate private rooms, where [group members were] "given instructions concerning the rules and payoffs of the relevant game. The instructions were phrased in neutral language (e.g., the pools were labelled ''A'' and ''B,'' with no reference to ''cooperation'' or ''competition'') and explained how each player's own decision and the decisions of the other players would affect the payoffs."

"Participants subsequently answered a short quiz that tested their understanding of the game's rules. Each participant received an endowment of 10 tokens and had to indicate on a ''decision form'' how many tokens he was keeping, how many he was contributing to pool W (the within group pool in the IPD-MD game), and how many he was contributing to pool B (the between-group pool in the IPD-MD game and the only pool in the IPD game). Each token that was contributed to either pool W or pool B paid [£0.20] to each ingroup member, including the contributor. Each token that was contributed to pool B also subtracted [£0.20] from the payoff to each outgroup member. For each token a player kept, he was paid [£0.40]. In addition, each player was paid a flat bonus of [£6] to ensure that payoffs would be positive. Thus, participants could earn between [£2] and [£14] depending on the outcome of the game."

"Participants in the no-communication condition made their decisions without communicating with the other ingroup members. Participants in the communication condition met with the other members of their team for 5min of free discussion and then returned to their rooms to make their decisions in private. Participants were explicitly assured that their decisions would remain confidential."

"After submitting their decisions, but before learning the outcomes of the game, participants responded to a post experimental questionnaire assessing their expectations regarding the behaviour of ingroup and outgroup members (i.e., how many chips, on average, ingroup and outgroup members had contributed to each of the pools)."

Differences from the Original Study

In the original study, participants were paid in NIS (new Israeli shekel). Because the replication's sample came from the University of Winchester in the UK, participants were paid in British Pound Sterling. On July 31, 2014 when the procedure was developed, one NIS was worth £0.17. For practical reasons, as there is a 20p (£0.2) coin, the £0.17 was rounded up to £0.20.

Additionally, the original study only included male participants. Because University of Winchester is heavily female dominated and has a small student body, collecting only male data would be much more difficult than including both sexes. A footnote in the original article indicated that in the IPD-MD game, there were "no differences between all-female and all-male groups" (Halevy et al., p. 407) suggesting that the inclusion of women in the study should not

affect the outcome. To avoid introducing new confounds, however, each pair of groups was required to be of the same sex.

Lastly, the original study issued instructions to participants in separate rooms. Due to space constraints, each group was placed in the same room, seated separately. In the no-communication condition, participants made their decisions without ever communicating with their group members. In the communication condition, group members drew together within their room to talk for 5 minutes.

Analysis Plan

As in the original study, all analyses were done at the group level.

The key effect identified by Dr. Halevy, $F(1, 76)=109.74$, $p < .001$, concerned contribution rates to pool B in the two games. The replicating authors planned to conduct a 2 (game type: IPD-MD vs. IPD) x 2 (communication type: none vs. within group) between subjects ANOVA on contribution rate to pool B. The effect of interest was the main effect of gain type; based on the original article it was anticipated that contributions to the IPD-MD game's pool B would be lower.

A supplementary analysis was run on the IPD-MD games to investigate the effect of communication on contributions to pool W. The original article reported that contributions to pool W increased after within group communication, as evidenced by a Welch's t-test, $t(29) = -2.756$, $p = .010$. Welch's t-tests are used when the homogeneity of variance assumption is not met. The replicators planned to use a Welch's t-test if their data also violated the homogeneity of variance assumption. Should variance be homogenous, however, a Student's t-test was planned to be run.

Results

A 2 (game type: IPD-MD vs. IPD) x 2 (communication type: none vs. within group) between subjects ANOVA on the contribution rate to Pool B revealed that the contribution rate to Pool B was significantly lower in the IPD-MD (15%, 95% CI: 5.31, 25.36) than the contribution rate to Pool B (the only pool) in the IPD game (52%, 95% CI: 41.48, 61.53; $F(1, 36) = 26.76$, $p < .001$, $\eta^2 = .426$). This finding is consistent with the finding reported by Halevy et al. (2008) and can be seen in Table 1.

Table 1. Results of the between-subjects ANOVA on the contribution rate to Pool B.

	df	*F*	*p*	η^2
Game Condition	1	26.76	.000	.426
Communication Condition	1	0.16	.688	.005
Game Condition * Communication Condition	1	0.21	.653	.006
Error	36	--	--	--

As the homogeneity of variance assumption was met, we further used a Student's t-test to investigate the effect of communication on contributions to Pool W. In contrast to the original article, communication did not impact on the allocation of tokens to pool W. The average contribution of tokens to Pool W was not significantly higher in the communication condition (49%) when compared to the non-communication condition (35%; $t(18) = -1.17$, $p = .254$). This finding is further detailed in Table 2.

Table 2. T-test results on the effect of communication on contributions to Pool W

	M	*SD*	*t*	*df*	*p*	*95% Confidence Interval of the Difference*	
No Communication	35.00	15.73				Upper	Lower
Withingroup communication	48.67	33.16	-1.18	18	.254	-38.05	10.72

Discussion

The present registered replication aimed to replicate two key findings reported in "'In-Group Love' and 'Out-Group Hate' as Motives for Individual Participation in Intergroup Conflict: A New Game Paradigm" by Halevy, Bornstein and Sagiv (2008, *Psychological Science*). The results of this replication are somewhat mixed but overall, we replicated the effect identified as the main finding by the original authors.

Our replication results indicate that participants contribute less tokens to the competitive Pool B in the IPD-MD condition than in the IPD condition. Thus, when given the choice to help their

ingroup while simultaneously harming the outgroup versus only helping their ingroup (IPD-MD), participants are more inclined to help their ingroup without harming to the outgroup than helping their ingroup and simultaneously harming the outgroup. If, however, participants are only given the option to simultaneously help their ingroup and harm the outgroup (IPD), they will do so. In contrast, testing whether communication altered the outcomes of participants' contributions to the cooperative Pool W did not yield any significant differences between IPD-MD groups who communicated with each other versus IPD-MD groups who did not engage in group discussions. This failure to replicate the effect of communication on the token allocations somewhat surprised us. However, looking at methodological differences between the original study and this replication, several possible explanations arise.

Contrary to the procedure reported by the original authors and due to space limitations at the replication site, we always tested participants in groups of three in the same lab. Even though we ensured that participants did not communicate with each other in the no communication conditions and always allocated their tokens in private, a sense of 'team spirit' may have developed due to the physical proximity with the ingroup. Furthermore, only eight of our experimental groups were male, while the remaining sample consisted of female participants. Although the original article reports no gender differences in token allocations in the IPD-MD game (Halevy et al., 2008; Footnote 2, p. 407), it is possible that this difference from the original research impacted on our replication findings. Finally, the original study was conducted in a different cultural context (Israel) than our replication (United Kingdom). It is therefore conceivable that actual cultural differences exist with reference to the impact of communication on token allocation.

The preparation and day to day data collection for this research was at times challenging. Replicating the original research as closely as possible turned out to be time consuming as well as demanding in terms of the organisation and administration of our and the participants' time in the lab. We definitely commend the original authors for collecting data from a sample twice the size of ours. Another difficulty was the standard of reporting in the original article. Even if we acknowledge *Psychological Science*'s preference for short and concise papers, without the support of Dr Halevy this replication would not have been possible from the information reported in the paper only. That said, Dr Halevy greatly assisted us with detailed and timely responses and clarifications to any queries.

This registered replication strengthens our understanding of people's motivations to engage in intergroup conflict. While the role of ingroup communication remains ambiguous, our evidence supports Halevy et al.'s (2008) finding that, given the choice, people engage in conflicts to help their own rather than harming others.

References

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